3.4 COMPARE TO OTHER EXISTING APPROACHES

We conduct a set of additional studies and show that self-consistency significantly outperforms existing methods including sample-and-rank, beam search, and ensemble-based approaches.

Comparison to Sample-and-Rank One commonly used approach to improve generation quality is sample-and-rank, where multiple sequences are sampled from the decoder and then ranked according to each sequence's log probability (Adiwardana et al., 2020). We compare self-consistency with sample-and-rank on GPT-3 *code-davinci-001*, by sampling the same number of sequences from the decoder as self-consistency and taking the final answer from the top-ranked sequence. The results are shown in Figure 3. While sample-and-rank does improve the accuracy with additionally sampled sequences and ranking, the gain is much smaller compared to self-consistency.

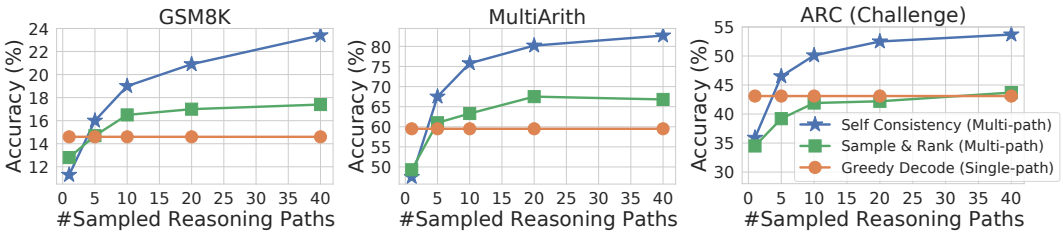

Figure 3: Self-consistency significantly outperforms sample-and-rank with the same # of samples.

Comparison to Beam Search In Table 6, we compare self-consistency with beam search decoding on the UL2-20B model. For a fair comparison we report the accuracy under the same number of beams and reasoning paths. On both tasks self-consistency outperforms beam search significantly. Note self-consistency can also adopt beam search to decode each reasoning path (results are shown as "Self-consistency using beam search"), but its performance is worse compared to self-consistency with sampling. The reason is that beam search yields a lower diversity in the outputs (Li & Jurafsky, 2016), while in self-consistency the diversity of the reasoning paths is the key to a better performance.

	Beam size / Self-consistency paths	1	5	10	20	40
AQuA	Beam search decoding (top beam)	23.6	19.3	16.1	15.0	10.2
	Self-consistency using beam search	23.6	19.8 $\pm$ 0.3	21.2 $\pm$ 0.7	24.6 $\pm$ 0.4	24.2 $\pm$ 0.5
	Self-consistency using sampling	19.7 $\pm$ 2.5	**24.9** $\pm$ **2.6**	**25.3** $\pm$ **1.8**	**26.7** $\pm$ **1.0**	**26.9** $\pm$ **0.5**
MultiArith	Beam search decoding (top beam)	10.7	12.0	11.3	11.0	10.5
	Self-consistency using beam search	10.7	11.8 $\pm$ 0.0	11.4 $\pm$ 0.1	12.3 $\pm$ 0.1	10.8 $\pm$ 0.1
	Self-consistency using sampling	9.5 $\pm$ 1.2	11.3 $\pm$ 1.2	**12.3** $\pm$ **0.8**	**13.7** $\pm$ **0.9**	**14.7** $\pm$ **0.3**

Table 6: Compare self-consistency with beam search decoding on the UL2-20B model.

Comparison to Ensemble-based Approaches We further compare self-consistency to ensemble-based methods for few-shot learning. In particular, we consider ensembling by: (1) prompt order permutation: we randomly permute the exemplars in the prompt 40 times to mitigate model's sensitivity to prompt order (Zhao et al., 2021; Lu et al., 2021); and (2) multiple sets of prompts (Gao et al., 2021): we manually write 3 different sets of prompts. We took majority vote of the answers from greedy decoding in both approaches as an ensemble. Table 7 shows that compared to self-consistency, existing ensemble-based approaches achieve a much smaller gain.[8] In addition, note that self-consistency is different from a typical model-ensemble approach, where *multiple* models are trained and their outputs are aggregated. Self-consistency acts more like a "self-ensemble" on top of a *single* language model. We additionally show the results of ensembling multiple models in Appendix A.1.3 where the model-ensembles perform much worse compared to self-consistency.

	GSM8K	MultiArith	SVAMP	ARC-e	ARC-c
CoT (Wei et al., 2022)	17.1	51.8	38.9	75.3	55.1
Ensemble (3 sets of prompts)	18.6 $\pm$ 0.5	57.1 $\pm$ 0.7	42.1 $\pm$ 0.6	76.6 $\pm$ 0.1	57.0 $\pm$ 0.2
Ensemble (40 prompt permutations)	19.2 $\pm$ 0.1	60.9 $\pm$ 0.2	42.7 $\pm$ 0.1	76.9 $\pm$ 0.1	57.0 $\pm$ 0.1
Self-Consistency (40 sampled paths)	**27.7** $\pm$ **0.2**	**75.7** $\pm$ **0.3**	**53.3** $\pm$ **0.2**	**79.3** $\pm$ **0.3**	**59.8** $\pm$ **0.2**

Table 7: Self-consistency outperforms prompt-order and multi-prompt ensembles on LaMDA-137B.

[8] Self-consistency is compatible with both ensemble approaches and we show the results in Appendix A.1.4.

3.5 ADDITIONAL STUDIES

We conducted a number of additional experiments to analyze different aspects of the self-consistency method, including its robustness to sampling strategies and parameters, and how it works with imperfect prompts and non-natural-language reasoning paths.

Self-Consistency is Robust to Sampling Strategies and Scaling We show self-consistency is robust to sampling strategies and parameters, by varying T in temperature sampling (Ackley et al., 1985; Ficler & Goldberg, 2017), k in top-k sampling (Fan et al., 2018; Holtzman et al., 2018; Radford et al., 2019), and p in nucleus sampling (Holtzman et al., 2020), over PaLM-540B in Figure 4 (left). Figure 4 (right) shows that self-consistency robustly improves performance across all scales for the LaMDA-137B model series. The gain is relatively lower for smaller models due to certain abilities (e.g., arithmetic) only emerge when the model reaches a sufficient scale (Brown et al., 2020).

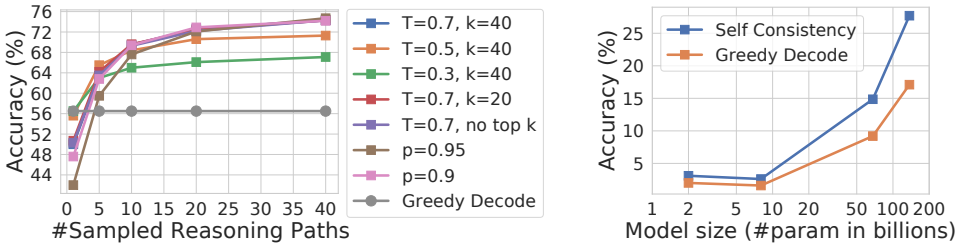

Figure 4: GSM8K accuracy. (Left) Self-consistency is robust to various sampling strategies and parameters. (Right) Self-consistency improves performance across language model scales.

Self-Consistency Improves Robustness to Imperfect Prompts For few-shot learning with manually constructed prompts, human annotators sometimes make minor mistakes when creating the prompts. We further study if self-consistency can help improve a language model's robustness to imperfect prompts.[9] We show the results in Table 8: while imperfect prompts decrease accuracy with greedy decoding (17.1 → 14.9), self-consistency can fill in the gaps and robustly improve the results.

Additionally, we found that the consistency (in terms of % of decodes agreeing with the final aggregated answer) is highly correlated with accuracy (Figure 5, over GSM8K). This suggests that one can use self-consistency to provide an *uncertainty estimate* of the model in its generated solutions. In other words, one can use low consistency as an indicator that the model has low confidence; i.e., self-consistency confers some ability for the model to "know when it doesn't know".

LaMDA-137B	Prompt with correct chain-of-thought	17.1
	Prompt with imperfect chain-of-thought	14.9
	+ Self-consistency (40 paths)	**23.4**
	Prompt with equations	5.0
	+ Self-consistency (40 paths)	**6.5**
PaLM-540B	Zero-shot CoT (Kojima et al., 2022)	43.0
	+ Self-consistency (40 paths)	**69.2**

Table 8: Self-consistency works under imperfect prompts, equation prompts and zero-shot chain-of-thought for GSM8K.

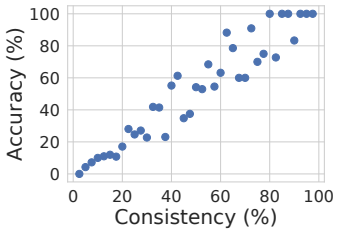

Figure 5: The consistency is correlated with model's accuracy.

Self-Consistency Works for Non-Natural-Language Reasoning Paths and Zero-shot CoT We also tested the generality of the self-consistency concept to alternative forms of intermediate reasoning like equations (e.g., from "*There are 3 cars in the parking lot already. 2 more arrive. Now there are 3 + 2 = 5 cars.*" to "*3 + 2 = 5*"). The results are shown in Table 8 ("Prompt with equations"): self-consistency still improves accuracy by generating intermediate equations; however, compared to generating natural language reasoning paths, the gain is smaller since the equations are much shorter and less opportunity remains for generating diversity in the decoding process. In addition, we tested self-consistency with zero-shot chain-of-thought (Kojima et al., 2022) and show that self-consistency works for zero-shot CoT as well and improves the results significantly (+26.2%) in Table 8.

[9] We use the same prompts as before, but swap all the numbers in the reasoning paths with random numbers except the final answer, e.g., from "*There are 3 cars in the parking lot already. 2 more arrive. Now there are 3 + 2 = 5 cars.*" to "*There are 7 cars in the parking lot already. 6 more arrive. Now there are 7 + 6 = 5 cars.*".

4 RELATED WORK

Reasoning in language models. Language models are known to struggle in Type 2 tasks, such as arithmetic, logical and commonsense reasoning (Evans, 2010). Previous work has primarily focused on *specialized* approaches for improving reasoning (Andor et al., 2019; Ran et al., 2019; Geva et al., 2020; Piękos et al., 2021). Compared to prior work, self-consistency is applicable to a wide range of reasoning tasks without any additional supervision or fine-tuning, while still substantially improving the performance of the chain-of-thought prompting approach proposed in Wei et al. (2022).

Sampling and re-ranking in language models. Multiple decoding strategies for language models have been proposed in the literature, e.g., temperature sampling (Ackley et al., 1985; Ficler & Goldberg, 2017), top-k sampling (Fan et al., 2018; Holtzman et al., 2018; Radford et al., 2019), nucleus sampling (Holtzman et al., 2020), minimum Bayes risk decoding (Eikema & Aziz, 2020; Shi et al., 2022), and typical decoding (Meister et al., 2022). Other work has sought to explicitly promote diversity in the decoding process (Batra et al., 2012; Li et al., 2016; Vijayakumar et al., 2018).

Re-ranking is another common approach to improve generation quality in language models (Adiwardana et al., 2020; Shen et al., 2021). Thoppilan et al. (2022) collect additional human annotations to train a re-ranker for response filtering. Cobbe et al. (2021) train a "verifier" to re-rank generated solutions, which substantially improves the solve rate on math tasks compared to just fine-tuning the language model. Elazar et al. (2021) improve the consistency of factual knowledge extraction by extending pre-training with an additional consistency loss. All these methods require either training an additional re-ranker or collecting additional human annotation, while self-consistency requires no additional training, fine-tuning, nor extra data collection.

Extract reasoning paths. Some previous work has considered task-specific approaches for identifying reasoning paths, such as constructing semantic graphs (Xu et al., 2021a), learning an RNN to retrieve reasoning paths over the Wikipedia graph (Asai et al., 2020), fine-tuning with human annotated reasoning paths on math problems (Cobbe et al., 2021), or training an extractor with heuristic-based pseudo reasoning paths (Chen et al., 2019). More recently, the importance of diversity in the reasoning processes has been noticed, but only leveraged via task-specific training, either through an additional QA model over extracted reasoning paths (Chen et al., 2019), or by the introduction of latent variables in a commonsense knowledge graph (Yu et al., 2022). Compared to these approaches, self-consistency is far simpler and requires no additional training. The approach we propose simply couples the generation of reasoning paths and a final answer by sampling from the decoder, using aggregation to recover the most consistent answer without additional modules.

Consistency in language models. Some prior work has shown that language models can suffer from inconsistency in conversation (Adiwardana et al., 2020), explanation generation (Camburu et al., 2020), and factual knowledge extraction (Elazar et al., 2021). Welleck et al. (2020) use "consistency" to refer to generating an infinite-length sequence in recurrent language models. Nye et al. (2021) improve the logical consistency of samples from a System 1 model by adding a System 2-inspired logical reasoning module. In this paper we focus on a slightly different notion of "consistency", i.e., utilizing answer consistency among diverse reasoning paths to improve accuracy.

5 CONCLUSION AND DISCUSSION

We introduced a simple yet effective method called self-consistency, and observed that it significantly improves accuracy in a range of arithmetic and commonsense reasoning tasks, across four large language models with varying scales. Beyond accuracy gains, self-consistency is also useful for collecting rationales when performing reasoning tasks with language models, and for providing uncertainty estimates and improved calibration of language model outputs.

One limitation of self-consistency is that it incurs more computation cost. In practice people can try a small number of paths (e.g., 5 or 10) as a starting point to realize most of the gains while not incurring too much cost, as in most cases the performance saturates quickly (Figure 2). As part of future work, one could use self-consistency to generate better supervised data to fine-tune the model, such that the model can give more accurate predictions in a single inference run after fine-tuning. In addition, we observed that language models can sometimes generate incorrect or nonsensical reasoning paths (e.g., the StrategyQA example in Table 4, the two population numbers are not exactly correct), and further work is needed to better ground models' rationale generations.